

- RAG Learning

6.2 Averages, Spread & Grouped Data

Pearson Edexcel GCSE Mathematics 1MA1 — Statistics — spec refs S3, S4 — Papers 1 (non-calculator), 2 and 3 (calculator)

In 1MA1, S4 is on both tiers, but the Higher version adds **box plots**, **quartiles** and the **inter-quartile range**. S3 (histograms with equal and unequal class widths, and cumulative frequency graphs) appears **only in the Higher content list**. Sections marked **Higher** are Higher tier only.

6.2.1 Types of Data (S4)

Discrete data — can only take certain values, usually found by *counting* (number of texts sent, shoe size).

Continuous data — can take any value within a range, found by *measuring* (mass, time, length).

Grouped data — values sorted into classes such as $10 < t \leq 20$, used when there are many different values.

Inequalities matter: in $10 < t \leq 20$, a time of exactly 10 belongs to the class *below*, and exactly 20 belongs in this class.

6.2.2 Mean, Median, Mode and Range (S4)

Mean = $\frac{\text{total of the values}}{\text{number of values}}$ Median = the $\left(\frac{n+1}{2}\right)$ th value once the data is in order

Mode = the most common value. **Range** = largest value — smallest value (a measure of

spread).

Worked Example: Data: 12, 7, 15, 9, 7, 18, 10, 14.

$$\text{Mean} = \frac{92}{8} = 11.5$$

In order: 7, 7, 9, 10, 12, 14, 15, 18. With $n = 8$ the median is the 4.5th value, halfway between 10 and 12, so median = 11.

$$\text{Mode} = 7. \quad \text{Range} = 18 - 7 = 11.$$

Working backwards from a mean

Use $\text{total} = \text{mean} \times \text{number of values}$.

Worked Example: The mean of five numbers is 16. A sixth number is added and the mean becomes 17. Work out the sixth number.

$$\text{Old total} = 5 \times 16 = 80. \quad \text{New total} = 6 \times 17 = 102. \quad \text{Sixth number} = 102 - 80 = 22.$$

Worked Example: 14 boys did a mean of 6.5 hours of homework; 16 girls did a mean of 8 hours. Work out the mean for all 30 pupils.

$$\frac{14 \times 6.5 + 16 \times 8}{30} = \frac{91 + 128}{30} = \frac{219}{30} = 7.3 \text{ hours.}$$

(Averaging the two means gives 7.25, which is wrong because the groups are different sizes.)

Outliers and choosing an average

An **outlier** is a value that is much bigger or smaller than the rest. It may be a mistake or a genuine extreme value. Outliers have a big effect on the **mean** and the **range**, but very little effect on the median (or the IQR).

Average	Best when...	Weakness
Mean	Numerical data without outliers; uses every value	Pulled towards outliers
Median	Data has outliers or is skewed (rents, salaries, house prices)	Ignores most of the values
Mode	Categorical data, or the most popular item is needed (sizes to stock)	May not exist, or there may be several

Worked Example: Monthly rents on a street: £650, £700, £720, £740, £780, £2400.

Mean = $5990 \div 6 = 998.33$, i.e. £998.33. Median = $\frac{720 + 740}{2} = 730$, i.e. £730.

The median is more typical: five of the six rents are below the mean, which has been dragged up by the outlier £2400.

6.2.3 Averages from a Frequency Table (S4)

$$\text{Mean} = \frac{\sum fx}{\sum f} = \frac{\text{total of (value} \times \text{frequency)}}{\text{total frequency}}$$

Worked Example: Goals scored by a team in 25 matches:

Goals x	Frequency f	fx	Running total
0	6	0	6
1	8	8	14
2	7	14	21

3	3	9	24
4	1	4	25
Total	25	35	

Mean = $35 \div 25 = 1.4$ goals. Mode = 1 goal.

Median = 13th value; the running total passes 13 in the "1" row (values 7 to 14), so median = 1 goal. Range = $4 - 0 = 4$.

Exam Tip: Divide by the **total frequency** (25), not the number of rows (5). The mode is the *value* with the highest frequency (1 goal), not the frequency itself (8).

6.2.4 Grouped Frequency Tables and Frequency Polygons (S4)

With grouped data the individual values are lost, so:

- the mean can only be **estimated**, using the **midpoint** of each class;
- we give the **modal class** (highest frequency) rather than a mode;
- we find the **class that contains the median** using running totals.

Worked Example: Times taken by 50 people to solve a puzzle:

Time t (min)	f	Midpoint x	fx	Running total
$0 < t \leq 4$	7	2	14	7
$4 < t \leq 8$	16	6	96	23
$8 < t \leq 12$	18	10	180	41
$12 < t \leq 16$	9	14	126	50

Total	50		416	
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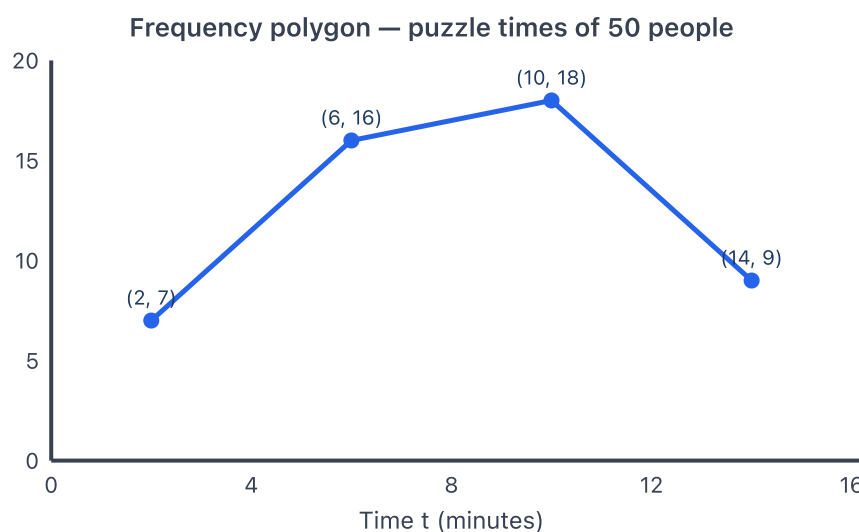
Estimated mean = $416 \div 50 = 8.32$ minutes.

Modal class: $8 < t \leq 12$. The median is the 25.5th value; running totals 23 then 41, so it lies in $8 < t \leq 12$.

Exam Tip: Sense-check: an estimated mean must lie between the smallest and largest possible values (0 and 16 here). Getting 104 means you divided by 4 classes instead of 50 people.

Frequency polygons

Plot each class frequency above the class **midpoint** and join the points with straight lines (do not join the first and last points to the axis unless told to). Two frequency polygons on the same axes are a quick way to compare distributions.



6.2.5 Comparing Distributions (S4)

A full comparison has **two** parts, each using numbers and written in context:

1. **Average** (mean or median): which group is higher or lower on average?

2. **Spread** (range, or IQR on Higher): which group is more consistent? A smaller spread means more consistent.

Worked Example: Morning group's quiz scores: 14, 18, 15, 20, 17, 16, 19. Afternoon group: median 15, range 11.

Morning: in order 14, 15, 16, 17, 18, 19, 20, so median = 17 and range = $20 - 14 = 6$.

The morning group had the higher median (17 vs 15), so they scored better on average.

The morning group had the smaller range (6 vs 11), so their scores were more consistent.

Exam Tip: Write comparisons like "Group A's median is higher, so on average Group A ..." and "Group B's range is smaller, so Group B's ... were more consistent." Be careful with context: for race times, a *lower* average means *faster*.

6.2.6 Quartiles and Inter-Quartile Range (S4) Higher

For ordered data with n values:

- **Lower quartile (LQ)** — the $\left(\frac{n+1}{4}\right)$ th value; a quarter of the data is below it.
- **Median** — the $\left(\frac{n+1}{2}\right)$ th value.
- **Upper quartile (UQ)** — the $\left(\frac{3(n+1)}{4}\right)$ th value; three quarters of the data is below it.

$$\text{IQR} = \text{UQ} - \text{LQ}$$

The IQR is the spread of the **middle half** of the data, so extreme values do not affect it.

Worked Example: Reaction times (hundredths of a second) of 11 students: 13, 15, 18, 20, 21, 24, 26, 27, 30, 33, 36.

$n + 1 = 12$: LQ = 3rd value = 18; median = 6th value = 24; UQ = 9th value = 30.
 IQR = $30 - 18 = 12$. Range = $36 - 13 = 23$.

Worked Example: Testing for outliers. A common rule: a value is an outlier if it is more than $1.5 \times \text{IQR}$ above the UQ or below the LQ.

Data: 2, 5, 6, 8, 9, 10, 12, 13, 14, 16, 31. LQ = 6, UQ = 14, IQR = 8, and $1.5 \times 8 = 12$.

Upper limit = $14 + 12 = 26$; lower limit = $6 - 12 = -6$. Since $31 > 26$, 31 is an outlier; no value is below -6 .

6.2.7 Cumulative Frequency Graphs (S3) Higher

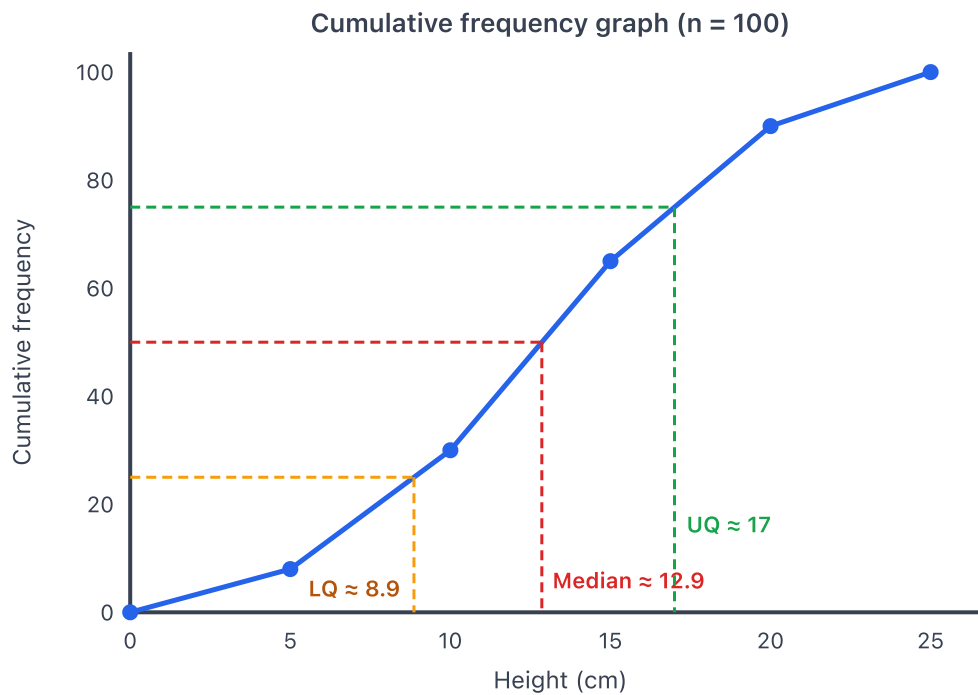
Cumulative frequency is a running total of the frequencies. Plot each cumulative frequency at the **upper class boundary**, start at the lowest boundary with cumulative frequency 0, and join the points with a smooth curve or straight lines.

With n values: **median** at $\frac{n}{2}$, **LQ** at $\frac{n}{4}$, **UQ** at $\frac{3n}{4}$ on the cumulative frequency axis.

Worked Example: Heights of 100 seedlings:

Height h (cm)	Frequency	Cumulative frequency	Point to plot
$0 < h \leq 5$	8	8	(5, 8)
$5 < h \leq 10$	22	30	(10, 30)
$10 < h \leq 15$	35	65	(15, 65)
$15 < h \leq 20$	25	90	(20, 90)
$20 < h \leq 25$	10	100	(25, 100)

Start the graph at (0, 0).



Reading the graph: $LQ \approx 8.9$ cm, median ≈ 12.9 cm, $UQ \approx 17$ cm, so $IQR \approx 17 - 8.9 = 8.1$ cm. (A curve drawn by hand gives slightly different readings; mark schemes accept a range.)

Worked Example: "How many are more than...?" Estimate how many seedlings are taller than 18 cm.

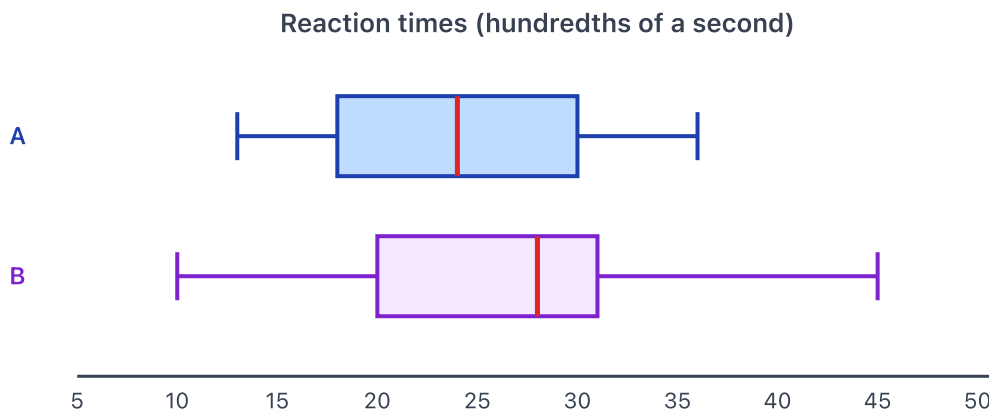
Read up from 18 cm: cumulative frequency ≈ 80 . So about $100 - 80 = 20$ seedlings are taller than 18 cm.

Exam Tip: Plot at the **upper** boundary, never the midpoint. Leave your reading lines on the graph — they earn method marks. For "more than" questions, subtract your reading from the total.

6.2.8 Box Plots (S4) Higher

A **box plot** is drawn on a scale and shows five values: **minimum**, **LQ**, **median**, **UQ**, **maximum**. The box goes from LQ to UQ (its length is the IQR) with a line at the median;

whiskers reach out to the minimum and maximum.



Worked Example: Group A (from 6.2.6): min 13, LQ 18, median 24, UQ 30, max 36. Group B: min 10, LQ 20, median 28, UQ 31, max 45.

IQR: $A = 30 - 18 = 12$; $B = 31 - 20 = 11$.

Average: A's median (24) is lower than B's (28), so Group A generally reacted **faster**.

Spread: B's IQR (11) is slightly smaller than A's (12), so the middle half of Group B was slightly **more consistent**, even though B's range (35) is much larger than A's (23).

Exam Tip: A box plot tells you proportions: 25% of values are below the LQ, 50% lie inside the box, 75% are below the UQ. It does *not* show the mean or how many values there are. When comparing, use the median and IQR rather than the range, since the range depends on just two (possibly extreme) values.

6.2.9 Histograms (S3) Higher

A **histogram** shows grouped continuous data on a continuous horizontal scale, with **no gaps** between bars. The **area** of each bar represents its frequency, so the vertical axis is **frequency density**. When all classes have the same width, bar heights are in proportion to the frequencies; with **unequal** widths you must use frequency density.

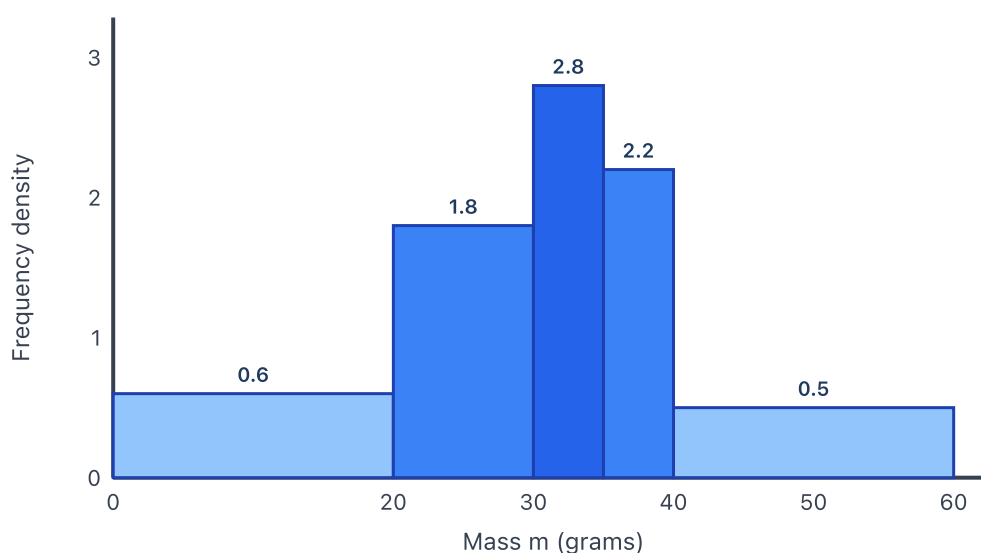
$$\text{Frequency density} = \frac{\text{frequency}}{\text{class width}}$$

$$\text{Frequency} = \text{frequency density} \times \text{class width}$$

Worked Example: Masses of 65 letters:

Mass m (g)	Frequency	Class width	Frequency density
$0 < m \leq 20$	12	20	$12 \div 20 = 0.6$
$20 < m \leq 30$	18	10	$18 \div 10 = 1.8$
$30 < m \leq 35$	14	5	$14 \div 5 = 2.8$
$35 < m \leq 40$	11	5	$11 \div 5 = 2.2$
$40 < m \leq 60$	10	20	$10 \div 20 = 0.5$

Histogram — masses of letters



Worked Example: Part of a class. Estimate how many letters have a mass between 25 g and 50 g.

25 to 30: $5 \times 1.8 = 9$; 30 to 40: $14 + 11 = 25$; 40 to 50: $10 \times 0.5 = 5$.

Estimate = $9 + 25 + 5 = 39$ letters (assuming the values are spread evenly within each class).

Worked Example: Finding the total. A histogram has bars 0–10 with FD 1.4, 10–25 with FD 2 and 25–45 with FD 0.65.

Frequencies: $10 \times 1.4 = 14$, $15 \times 2 = 30$, $20 \times 0.65 = 13$. Total = 57.

Exam Tip: If the frequency density axis has no numbers, use a class whose frequency you know to find the scale (frequency $\div$ width = FD for that bar's height). Label the axis "Frequency density", never "Frequency".

Choosing a diagram for grouped data (S3)

Diagram	Use it to...
Frequency polygon	Show the shape of grouped data and compare two distributions on one grid
Histogram Higher	Display continuous data, especially with unequal class widths
Cumulative frequency graph Higher	Estimate the median, quartiles and "how many above/below" a value
Box plot Higher	Compare medians and IQRs of two or more data sets at a glance

Summary

Measure / diagram	Key fact
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Mean	Total $\div n$; from a table $\sum fx \div \sum f$; grouped data uses midpoints (an estimate)
Median	$\left(\frac{n+1}{2}\right)$ th value in order
Mode / modal class	Most frequent value / class
Range	Largest — smallest; badly affected by outliers
Comparing	One average and one spread, both in context
Quartiles, IQR Higher	LQ $\left(\frac{n+1}{4}\right)$ th, UQ $\left(\frac{3(n+1)}{4}\right)$ th; IQR = UQ — LQ
Cumulative frequency Higher	Plot at upper class boundaries; read at $\frac{n}{4}, \frac{n}{2}, \frac{3n}{4}$
Box plot Higher	Min, LQ, median, UQ, max
Histogram Higher	FD = frequency $\div$ class width; area = frequency